

Problem

Given a value V , you have an infinite supply of N different coin denominations. Find the minimum number of coins required to make exactly V cents.

You must also print the coins used in an optimal solution.

If it is not possible to make the value V , print -1.

Input Format

The first line contains an integer T , the number of test cases.

For each test case: The first line contains two integers V and N .

V — the target value. N — the number of available coin denominations. The second line contains N integers $C[i]$, representing the coin denominations.

Constraints

$1 \leq T \leq 100$ $1 \leq V \leq 10^6$ $1 \leq N \leq 10^6$ $1 \leq C[i] \leq 10^6$

Output Format

For each test case:

Print the coins used in a minimum-coin solution, separated by spaces. In the next line, print the minimum number of coins. If it is impossible to make V , print -1.

Submissions

Leaderboard

Discussions

Sample Input 0

7 2
2 4

Sample Output 0

-1

Sample Input 1

11 4
1 5 6 9

Sample Output 1

2

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Language Java 15



```
1 import java.util.*;
2
3 class Solution {
4
5     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8
9         int V = sc.nextInt();
10        int N = sc.nextInt();
11
12        int[] coins = new int[N];
13
14        for (int i = 0; i < N; i++) {
15            coins[i] = sc.nextInt();
16        }
17
18        int[] dp = new int[V + 1];
19
20        Arrays.fill(dp, V + 1);
21
22        dp[0] = 0;
23
24        for (int amount = 1; amount <= V; amou
25
```

Line: 43 Col: 2



Upload Code as File

Run Code

Submit Code



Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?



Test case 0



Test case 1

Compiler Message

Success